

1 VEHICLE MOTION

Below, let m (kg) be the mass of a vehicle, v m/s be its speed and $g \approx 9.8 \text{ m/s}^2$ be the gravitational acceleration. The product mg (N) is the weight of the vehicle, where (N) indicates Newtons (the metric unit of force).

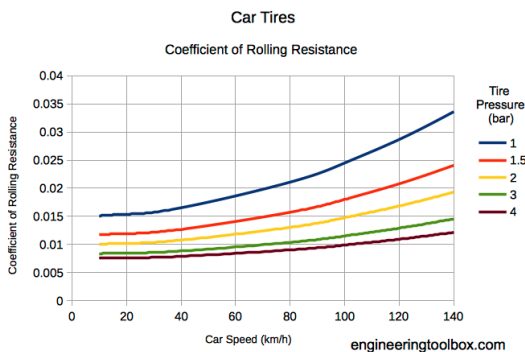
This class is concerned with four forces: the propulsive force from the vehicle itself, rolling resistance, aerodynamic resistance (drag), and grade resistance (gravity).

2 ROLLING RESISTANCE

Pushes in the opposite direction of the vehicle's motion. It is caused mainly because of the deformation of the tires: some of the energy used to compress the tires as they roll along the ground is dissipated as heat. Rolling resistance force is (approximately) the product of weight and a constant f_r :

$$F_r = f_r mg, \quad (\text{N}) \quad (1)$$

where f_r is the "coefficient of rolling resistance." f_r depends on road quality, how new the tires are, whether the tires are inflated and (most importantly to us) *speed*. For example ...



There are lots of studies on f_r , and most give an equation like

$$f_r = f_{r,0} + f_{r,1}v + f_{r,2}v^2 + \dots$$

where $f_{r,0}$, $f_{r,1}$, etc. are constants that you multiply by some power of v and depend on tire pressure etc.

The important thing is that f_r increases with speed. For our problems, we'll use a first-order approximation for f_r :

$$f_r = \frac{1}{100} + \frac{v}{4473} = 0.01 \cdot \left(1 + \frac{v}{44.73}\right). \quad (2)$$

so that

$$F_r = f_r \cdot (mg) = 0.01 \cdot \left(1 + \frac{v}{44.73}\right) \cdot (mg) \quad (\text{N}) \quad (3)$$

3 AERODYNAMIC RESISTANCE (DRAG)

Drag pushes in the opposite direction of the vehicle's motion, as the vehicle has to push air out of the way. Drag is given by

$$F_a = .5\rho C_D A v^2, \quad (\text{N}) \quad (4)$$

where ρ (kg/m³) is the density of air, C_D is the coefficient of drag (dimensionless), A (m²) is the frontal area of the vehicle.

4 GRADE RESISTANCE (GRAVITY)

Pushes with or against the vehicle's motion, depending on whether the vehicle is headed downhill (negative grade, pushes the vehicle forward) or uphill (positive grade, pulls vehicle backward).

$$F_g \approx (mg) \cdot G, \quad (\text{N}) \quad (5)$$

where $G = dy/dx$ is the grade of the road (e.g., $G=.015$).

5 PROPULSIVE FORCE

The force created by the tires pushing the vehicle forward along the road.

$$P = F_p v, \quad (\text{N}) \quad (6)$$

where P is the "power" (force times speed) generated by the engine and can be measured in horsepower or watts (W). One horsepower is 745.7 Watts.

NOTE ON ACCELERATION

If we know the forces on a vehicle, and its mass, then we can find its acceleration by the old $F = ma$ relation

$$a = (F_p - F_a - F_r - F_g)/m \quad (\text{m/s}^2). \quad (7)$$

If the vehicle is at constant speed, $a = 0$, which requires that the term in parentheses above must be zero. In that case, the forces pushing the vehicle forward and backward must be equal. Thus,

$$F_p = F_a + F_r + F_g. \quad (\text{N}) \quad (8)$$

By our definition of power, the power being expended in this case must be

$$P = F_p \cdot v = (F_a + F_r + F_g) \cdot v. \quad (9)$$

Since F_a varies with v^2 , this means P is a cubic polynomial of v (i.e., one with a v^3 in it).

6 STOPPING SIGHT DISTANCE

The main topic of driver characteristics we're concerned with is **stopping sight distance**: the distance a driver needs to stop. The *most basic* formula is

$$x_{SSD} = t_{pr}v_0 + \frac{v_0^2}{2a}, \quad (10)$$

where

- v_0 (m/s): initial speed
- t_{pr} (s): perception-reaction time. The time the driver wastes before applying the brakes. The AASHTO¹ recommends $t_{pr} = 2.5$ s, but we use different ones in the problems to emphasize it's a choice of the roads agency—not some physical law. For example, if everyone keeps texting while driving more and more then we'll have to use a higher t_{pr} .
- $t_{pr}v_0$ (m): the distance traveled during the perception-reaction time
- a (m/s²): the braking deceleration. (Below we'll break this out into two parts.)
- $v_0^2/2a$ (m): the braking distance. How far the vehicle travels while braking.

In turn, our deceleration term, a , can be broken out as the sum:

$$a = a_b + gG, \quad (11)$$

where

- G is the grade expressed as a decimal like 3% grade $\Rightarrow G = 0.03$
- g is gravitational acceleration
- gG is the resulting acceleration of gravity in the direction of travel, taking into account the grade (i.e., on a flat road where $G = 0$ then gravity makes no real difference to deceleration)
- a_b is the ordinary braking deceleration—what is comfortable for drivers when they are on a flat surface. AASHTO gives 3.4 m/s².

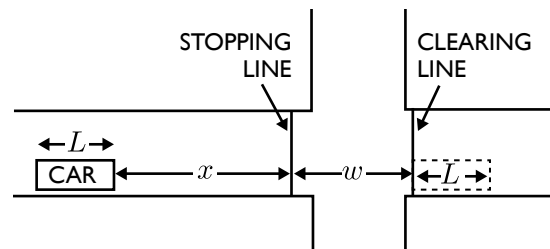
¹American Association of State Highway and Transportation Officials

Plugging (11) into (10) gives the equation we'll use most often in class:

$$x_{SSD} = t_{pr}v_0 + \frac{v_0^2}{2(a_b + gG)}. \quad (12)$$

7 YELLOW LIGHT DILEMMA ZONES

Consider the figure below. A car approaches an intersection moving west→east. When the light turns yellow, the car is x (m) from the stopping line and moving with speed v_0 (m/s). The yellow light lasts t_y (sec). The driver faces a choice of whether to clear the intersection or stop.



The driver can clear the intersection before the light turns red (without speeding up) only if

$$v_0 \cdot t_y \geq x + w + L, \quad (\text{can clear}) \quad (13)$$

where $v_0 t_y$ is the distance the car can travel during the yellow. Rearranging, the condition is

$$x \leq t_y v_0 - w - L. \quad (\text{can clear}) \quad (14)$$

Call the right-hand side of this equation x_{cl} , the **clearing distance**:

$$x_{cl} := t_y v_0 - w - L. \quad (15)$$

So if $x \leq x_{cl}$, then the driver can clear. Otherwise, the driver will be in the intersection when during the red if they try to clear. Meanwhile, *the driver can only stop before the stop line if*

$$x_{SSD} \leq x \quad (\text{can stop}), \quad (16)$$

that is, if he/she is *further away* from the stopping line than x_{SSD} .

Now, suppose the driver is at a location x_0 such that $x_{cl} < x_0 < x_{SSD}$: **too close to stop and too far to clear**. The driver is in a catch 22.

This can only happen if $x_{SSD} > x_{cl}$... that is, if the stopping sight distance exceeds the clearance distance. In that case, we call the interval (x_{cl}, x_{SSD}) the **Dilemma Zone**. Figure 1 illustrates what this

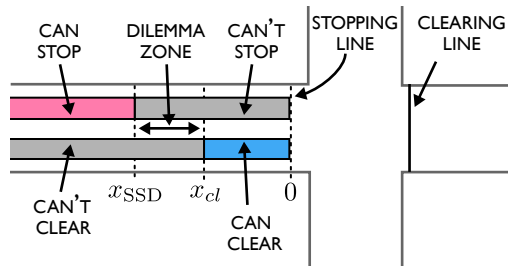
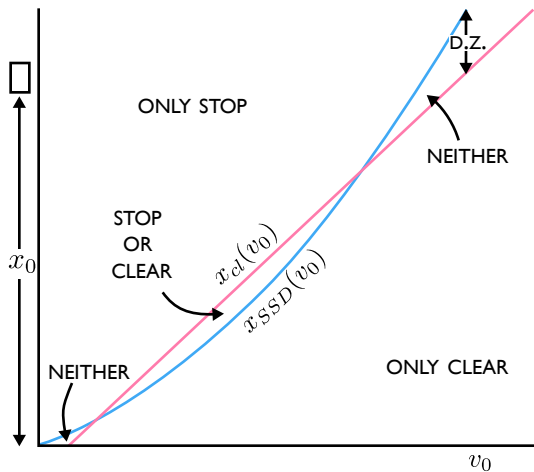


Figure 1: Dilemma Zone illustrated

Figure 2: x_{SSD} and x_{cl} plotted for different speeds

looks like in the intersection. A good way to analyze the intersection design variables is to plot x_{SSD} and x_{cl} as functions of speed, as in Fig. 2. The vertical axis indicates a car's distance from the stopping line when the light turns yellow, the horizontal axis the car's initial speed. Different regions indicate what the driver's options are for the v_0/x_0 combination suggested by the point. In the figure, a dilemma zone exists whenever the blue line $x_{SSD}(v_0)$ exceeds the pink line $x_{cl}(v_0)$. A dilemma zone for one speed is labeled "D.Z." If a dilemma zone exists for the design speed you've chosen for the road, then you might want to change the design speed or the yellow light time t_y .

8 GEOMETRY OF HORIZONTAL CURVES

Horizontal curves connect two pieces of road moving at different angles as you would view it on a map—from above. We focus on "simple" horizontal curves which are arcs of circles. Fig. 3 labels names for pieces involved in describing horizontal curves.

- PC = Point of Curve (beginning)

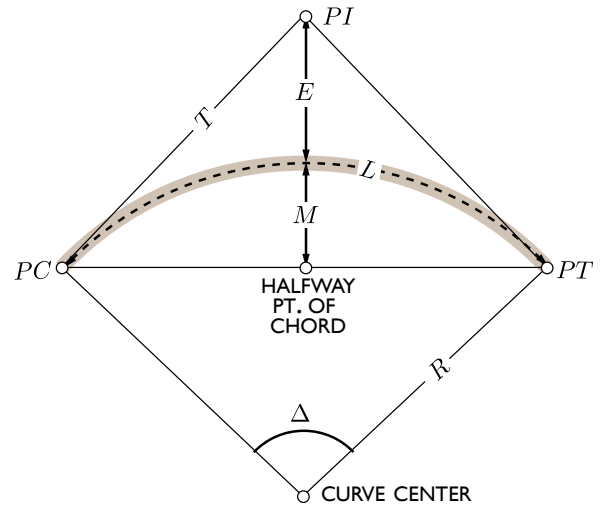


Figure 3: Elements of horizontal curve

- PT = Point of Tangent (ending)
- PI = Point of Intersection
- Δ = central angle ($^\circ$)
- R radius of curve (m)
- L length of curve (m)
- T tangent length (m)
- E external distance (m)
- M middle ordinate (m)

Here are equations for the distances:

$$L = (\pi/180) \cdot R\Delta \quad (17)$$

$$M = R \cdot [1 - \cos(\Delta/2)] \quad (18)$$

$$T = R \cdot \tan(\Delta/2) \quad (19)$$

$$E = R \cdot \left[\frac{1}{\cos(\Delta/2)} - 1 \right] \quad (20)$$

9 STOPPING SIGHT DISTANCE

For horizontal curves, it is easy to see around to the other side of the curve...unless there is an obstruction— e.g., rocks, a hill, trees, buildings, a sign, etc.

Consider the situation in Figure 4. The first car sees the second car just time to stop (the horizontal line is the line of sight). In the process of stopping, the first car traces out an arc (shown as a black dashed line) with a length equal to the stopping sight distance. Just as the horizontal curve itself forms a piece of a circle, this little arc is part of *another* circle.

This second circle has the same *center* as the origi-

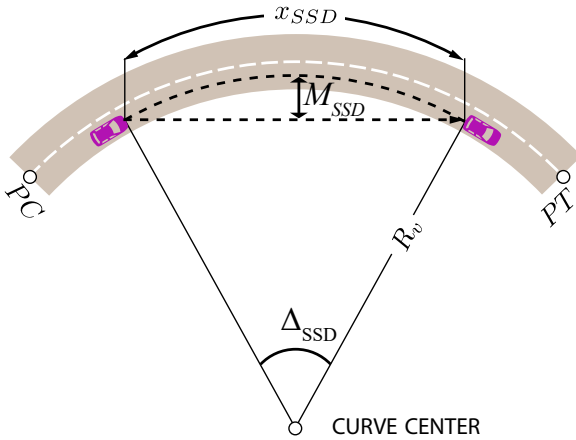


Figure 4: Horizontal curve and sight distance

nal but a smaller radius of R_v : this is the distance to the path traveled by a vehicle in the inner-most lane of the road. R_v is shorter than R , because R is measured to the *center of the road*. The little arc (the black dashed arc) also has its own central angle Δ_{SSD} , its own middle ordinate M_{SSD} , and its own curve length x_{SSD} . These quantities are all related just like the quantities of the overall curve.

$$M_{SSD} = R_v \cdot \left[1 - \cos\left(\frac{90x_{SSD}}{R_v\pi}\right) \right]. \quad (21)$$

Why do we care about the middle ordinate of the little arc? M_{SSD} is the **distance there needs to be from the center of the inner-most lane to any sight obstructions**. If there are bushes or signs located closer to the center of the inner-most lane than M_{SSD} , the driver will not be able to see a hazard in time to stop.

10 VEHICLE CORNERING

Consider a car going around a curve as in the diagram.

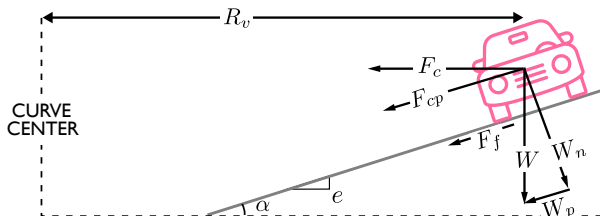


Figure 5: Cornering

- R_v = radius of vehicle's traveled path (m)
- α = angle of incline ($^\circ$)

- $e = \tan(\alpha)$ superelevation (%)
- W vehicle weight (N)
- W_n = weight *normal* to road (N)
- W_p = weight *parallel* to road (N)
- F_c = centripetal force (N)
- F_{cp} = centripetal force *parallel* to road (N)
- F_f = Frictional side force (N)

A car traveling a horizontal curve accelerates in a new direction (in the figure, to the left). This is why you lean when you turn on a curve. For a car with mass m to travel a curve with radius R_v (m) at speed v , the centripetal force on the car must be

$$F_c = \frac{Wv^2}{gR_v}, \quad (\text{N}) \quad (22)$$

where g is gravitational acceleration.

This force gets imparted to the car in two ways: gravity and "side friction." Gravity acts on the car because the superelevation (the sideways tilt) of the roadway pulls the car down and toward the center of the curve; this is the weight W_p in the figure. Side friction is the force of the roadway pushing on the tires toward the center of the curve; this is the force F_f in the figure.

Suppose we want to find the minimum R_v needed to balance the forces on the car as it turns at a speed of v , which we'll call $R_v^{\min}$. Using some simple physics, we can derive the relationship:

$$R_v^{\min} = \frac{v^2}{g \cdot (f_s + e)}. \quad (23)$$

where f_s is the **coefficient of side friction**. f_s depends on the tires, the road surface, etc. To check if the curve is safe, you check if the actual R_v is larger than this $R_v^{\min}$.

Alternatively, suppose that you want to check if the superelevation is adequate for a given R_v . To do so, replace $R_v^{\min}$ in equation (23) with R_v and replace e with $e^{\min}$. Then solve for $e^{\min}$:

$$e^{\min} = \frac{v^2}{R_v \cdot g} - f_s \quad (24)$$

11 4-STAGE PLANNING

As the name indicates, four-stage transportation demand analysis takes place in four stages:

- Trip generation

- (ii) Trip distribution
- (iii) Mode choice
- (iv) Trip assignment/network equilibrium

12 LOGIT MODELS

Logit models are used in the **Mode Choice** stage to figure out the probability someone chooses each mode in their **choice set**. A choice set might be: (car, bus, bike, metro). In this class, we focus on **binary** choices between *two* alternatives, such as between biking and walking.

The way this works is that you use a statistical model to calculate the **systematic utilities** of each mode (travel option). Suppose that a decision-maker chooses between alternative **A** and alternative **B**. You calculate that **A**'s systematic utility is V_A , and **B**'s systematic utility is V_B . In this case, the probability the decision-maker chooses **A** is

$$P_A = \frac{e^{V_A}}{e^{V_A} + e^{V_B}}. \quad (25)$$

13 TRIP ASSIGNMENT/NETWORK EQUILIBRIUM

The last stage in the 4-stage model determines what routes people take. This is especially important regarding car trips, because there is feedback between people's choices and travel times: the more people take a certain route, the higher the travel times, which makes fewer people want to take that route.

Unlike in traffic flow theory, trip assignment treats travel time as a function of *flow*, q , rather than density. The most famous type of travel time/flow function is called a BPR (Bureau of Public Roads) function. It looks like this:

$$t(q) = t_f + \left(\frac{q}{\theta}\right)^\gamma \quad (26)$$

where γ and θ are parameters related to how sensitive travel time is to flow.

We only deal with a simple situation: when there are two routes, 1 and 2, from an **origin** to a **destination**. In this case, the routes have their own BPR functions $t_1(q_1)$ and $t_2(q_2)$. There is a total flow of Q on both routes, so

$$q_1 + q_2 = Q. \quad (27)$$

13.1 USER EQUILIBRIUM

The first thing we want to figure out is: how many people wind up taking each route?

To find out, we use **Wardrop's First Principle**: *the travel times on all routes people actually take are the same*. When we find q_1 and q_2 that satisfy this, we have a *user equilibrium*.

To find the user equilibrium, since $q_2 = Q - q_1$, we try

$$t_1(q_1) = t_2(Q - q_1) \quad (28)$$

and solve for q_1 . If we can find a value of q_1 which (i) is between 0 and Q ; and (ii) which solves which this equality, then we have the equilibrium condition. If there is no such value, then all travelers will wind up taking one route or the other.

13.2 SOCIAL OPTIMUM

The user equilibrium is not necessarily optimal, in the sense it might not minimize the total amount of time vehicles spending traveling. The total or aggregate travel time is

$$T = q_1 t_1(q_1) + q_2 t_2(q_2), \quad (29)$$

which we can write as a function of q_1 like

$$T(q_1) = q_1 t_1(q_1) + (Q - q_1) t_2(Q - q_1). \quad (30)$$

To find the social optimum value of q_1 , q_1^* , we take the derivative of T and set it equal to zero:

$$T'(q_1^*) = 0. \quad (31)$$

Then we solve for q_1^* .

13.3 PARADOXES

We discuss two paradoxes.

Pigou-Downs-Knight Paradox: When **expanding** a link leads people to shift links but does not reduce equilibrium travel time.

Braess Paradox: When **adding** a link makes the equilibrium travel time worse.

14 TRANSPORTATION ECONOMICS

The only thing we learn from transportation economics is a *static traffic model* with a single link from an origin to a destination. The link has a link performance function $T(q)$. Demand for trips on the link is given by a function $D(t, p)$, where t is the travel time and p is the toll. Given a toll p , equilibrium arises at a flow q_e such that

$$D[T(q_e), p] = q_e. \quad (32)$$

Or, equivalently, equilibrium arises at a travel time t_e such that

$$T[D(t_e, p)] = t_e. \quad (33)$$